

Jack Brett

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Experience

Postdoctoral Researcher and Lead Developer in Locative Games

August 2023 – Present

Bournemouth University - LoGaCulture (EU-Funded)

Poole, UK

- Lead researcher and developer on a [EU-funded imitative](#): designed and built gameplay systems, mechanics, and total-conversion plug-ins in **Unity (C#/C++)**. Shipped twenty games on **iOS** and **Android**, including networked multiplayer features and server integration.
- Architected [LUTE](#), an authoring toolkit that lets designers and non-programmers build MR, locative game experiences. This included editor tools, plug-in development, the required use of extensible OOP with SOLID principles on complex Unity codebases (published [multiple papers](#) for the project).
- Profiled and optimised for low-end mobile hardware in complex outdoor conditions; iterated through structured play-testing (RITE process) with delivery management via GitHub issues/release; worked with a group of designers from highly varied backgrounds.

Research Engineer and Doctoral Student

October 2019 – December 2023

ROLI & Bournemouth University, Centre for Digital Entertainment

London, UK

- Embedded within an [industry R&D team](#) building real-time interactive applications across Unity, UE4, Blender, C#, C++ and Typescript; team workflow was managed in Jira.
- Built training games with ROLI's live learning ecosystem (subscription model, progression and engagement loops); gained extensive exposure to live-service product patterns.
- Conducted empirical research at the intersection of education and interactive technology, designing and leading four user studies that resulted in a series of peer-reviewed conference and journal publications.

Lead Research Developer

December 2017 – July 2022

Vitae VR & King's College London

Bournemouth, UK

- Collaborated with the KCL Psychology Department and a clinical research team on [VStore](#), a VR functional-cognition assessment designed to improve the ecological validity of cognitive testing for dementia and Alzheimer's.
- Acted as independent consultant and developer across an academic-clinical-industry partnership, leading key aspects of the build and the transfer of the platform across hardware/software (PC, PSVR, web APIs).
- Technical: C++, UE4 (Blueprint Visual Scripting), Python (for data visualisation/API work), Android and git.

Selected Works

Place Maps | *HTML, JavaScript, TypeScript, Git, Photoshop*

2025 – Present

- [Place Maps](#) is a map of a landscape with clickable places, waymarked walks, live data sources and in-app foot routing (currently still in the [prototype](#) stage and welcome to contributions).
- The idea came to me during the LoGaCulture project; I found that after leaving the many towns and cities I visited, I never really found out much about them. Whilst most maps are built on routing, this map is built on **places**.

Scale Loop | *C#, shaderLABS, Unity, C++, Python, Photoshop, Git*

2021 – 2026

- Developed a synthesiser application using Unity where you can play in different musical scales from cultures around the world and record loops using different sounds
- Currently released on the [Microsoft Store](#) and [Google Play Store](#).

Education

Bournemouth University

Poole, UK

Engineering Doctorate alongside [ROLI](#)

2019 – 2023

Bournemouth University

Poole, UK

Games Technology (First Class Honours BSc)

2014 – 2017

Selected Publications

[LUTE: A Hypertextual Mixed Reality Game Engine](#) (ACM Hypertext '25)

[Understanding Attitudes on Mixed Reality Heritage](#) (NHT'25 Workshop at ACM Hypertext 2025)

[Authoring Tools for Mixed Reality](#) (NHT'23 Workshop at ACM Hypertext 2023)

[Graveyard Symphony: A Study Investigating the Effectiveness of a Prototype Music Game](#) (Games: Research and Practice)